HO CHI MINH CITY – UNIVERSITY OF TECHNOLOGY

LAB 9 REPORT

MEASURING FOCAL LENGTHS OF CONVERGENT AND DIVERGENT LENSES

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Subject: General Physics Labs
Class: CC02
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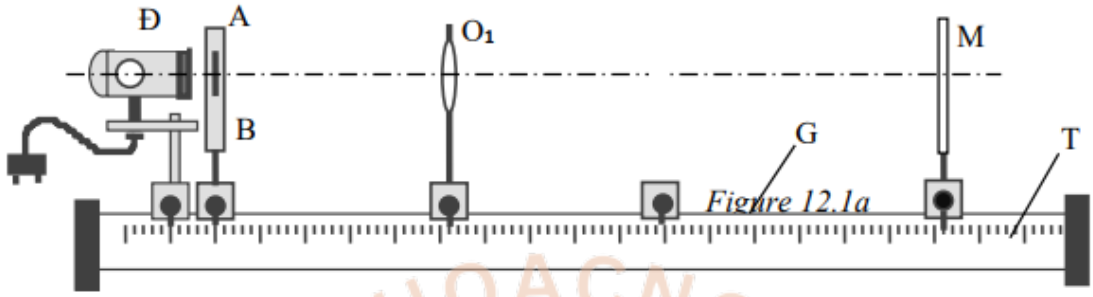

Figure .1

I/ OBJECTIVE

1. Examine the image forming through the convergent lens. Measure its focal length. 2. Examine the image forming through the lens system consisting of a convergent lens and a divergent len.
3. Measure the focal length of divergent lens.

II/ THEORY

The relation of the focal length f of the convex lens with distance d and d / from optical center of the lens to the AB object and A /B / image of AB can be written as:

$$\frac{1}{f} = \frac{1}{d} + \frac{1}{d\prime}$$

Therefore, we have:

$$\square = \frac{dd'}{d + d'}$$

Formula (12.1) and (12.2) are permutability with d and d' . In this experiment, we simultaneously measure the focal length of the convergent lens len O_1 and divergent lens O_2 using the optical bench.

III/ APPARATUS

- 1 optical bench 1000mm
- 1 convergent len O_1
- 1 divergent len O_2
- 1 light source Đ type 6V-8W
- 1 power source 6V-3A
- 1 AB object shape likes number “1” inside a circle hole of the plastic plate (H.1).
- 1 screen M 70x100mm

IV/ PROCEDURE

1. ***Measuring the focal length of convergent lens*** :

First method : Silberman method

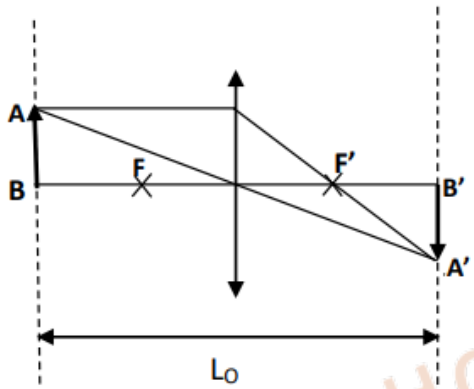

Figure .2

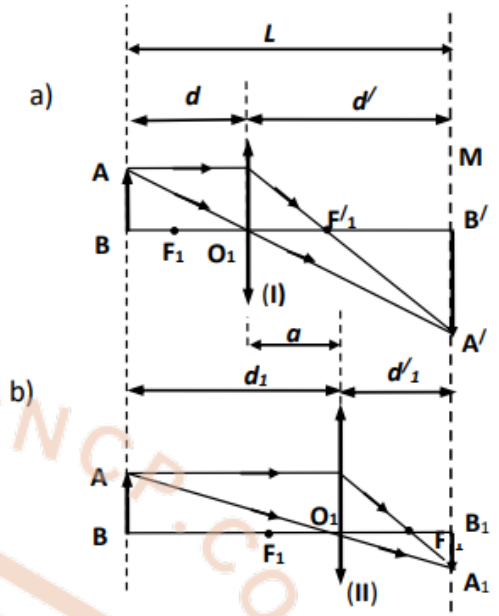

Figure. 3

a) Place AB near the light Đ at the mark 10cm. Next align the AB object in order to receive the homogeneous exposure. Place the AB and the screen M with the mutual distance less than 4f and the convergent lens is located between them.

b) Moving the convergent lens O_1 and the screen M so that the lens O_1 have the same distances with AB and M until obtaining the sharp image. In this configuration, the image has the same size as the object. (Moving the lens for a centimeter and the screen doubles the lens' moving distance until being near the sharpest state, moving with millimeter scale to have exact position). Write down the distance L_0 between AB and the screen M to table 1 in the report.

c) Make replications of the measurement in (b) triple times In this case, focal length f_1 of the convergent lens O_1 can be determined via:

$$f_1 = \frac{L_0}{4} \qquad (12.3)$$

Second method : Bessel method

a) Place the screen M far away AB with distance L 4. f on the optical bench (in this experiment, the suggested values are L= $4{,}5f_1$, L= $4{,}7f_1$, L= $4{,}9f_1$)

b) Moving the convergent lens O_1 from near AB object to farther away to position (I) at which the sharpest image $A'B'$ is larger than the AB object on the screen M (Figure.3a). Write down the coordinate x_1 of the lens O_1 at position (I) to the table 1 of the report.

c) Continue moving the convergent lens O_1 farther away AB to position (II) to obtain the sharpest image A_1B_1 smaller than AB on the screen M (Figure 3b). Write down the coordinate x_2 of the lens O_1 at position (II) to table 1 of the report.

d) Make replication of the task (b) và (c) triple times. In the above, the displacement a of the lens O_1 from position (I) to position (II) is:

$$\square = x_2 - x_1$$

and the focal length f_1 of the convergent lens O_1 can be determined by:

$$f_{1_i} = \frac{L_i^2 - a_i^2}{4L_i}$$

2. ***Measuring the focal length of the divergent lens: the connecting point method***

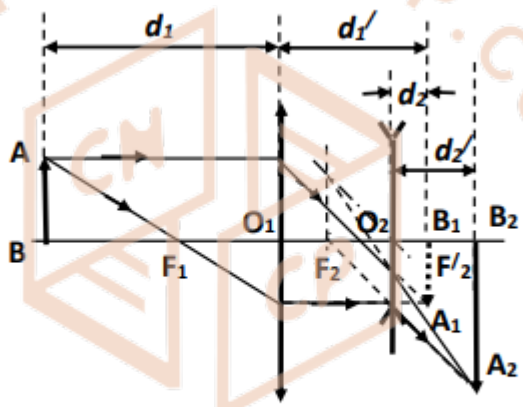

Figure .4

Divergent lenses only provide the real image of the imaginary object. Therefore to measure the focal length f_2 of the divergent lens O_2, we need to couple it with another convergent lens O_1 to be a coaxial lens system so that the real image A_1B_1 of AB object given by the convergent lens O_1 located back within the focal length $|f_2|$ of the divergent lens O_2 (Figure. 4) so the image A_1B_1 become the imagine object of the lens O_2 by follows:

a) Keeping position of AB object and the convergent lens O_1 at the position (II) in order to obtain the sharpest image A_1B_1 smaller than AB on the screen M in Figure 12.3b. Place the divergent lens O_2 on the track 4 at the back of the convergent lens O_1 and being coaxial with lens O_1, apart from the screen M distance $|d_2| = O_2B_1 < |f_2|$ (in this experiment, the suggested values are $|d_2|$ = 50mm,55mm,60mmm).

b) Moving the screen M farther away the divergent lens O_2 to the position $M^/$ to obtain the sharp image A_2B_2 apart from divergent lens O_2 distance $d^/_2$ in Figure

12.4. Make replication of this task triple times and write down the distance d'_2 in every time with the corresponding value of d_2 into the table 12.2 of the report.
Focal length f_2 of the divergent lens O_2 can be determined by:

$$f_2 = \frac{d_2 \ . d''_2}{d_2 + d'_2}$$

V/ EQUATIONS

- **Silberman's method:**

$$f_{1_i} = \frac{L_{o_i}}{4}$$

$$\Delta f_{1\,sys} = \frac{\Delta L_{o\,sys}}{4} \text{ (mm)}$$

$$\Delta f_1 = \underline{\Delta f_1} + \Delta f_{1\,sys} \text{ (mm)}$$

- **Bessel's method:**

$$f_{1_i} = \frac{{L_i}^2 - {a_i}^2}{4L_i} \text{ (mm)}$$

$$\frac{\Delta f_{1\,sys}}{f_{1\,mid}} = \left|\frac{2L_{mid}}{L_{mid}^2 - a_{mid}^2} - \frac{1}{L_{mid}}\right| \Delta L_{sys} + \left|\frac{-2a_{mid}}{L_{mid}^2 - a_{mid}^2}\right| \Delta a_{sys} \text{ (mm)}$$

$$\Delta f_1 = \underline{\Delta f_1} + \Delta f_{1\,sys} \text{ (mm)}$$

- **Connecting point method:**

$$f_{2_i} = \frac{d_{2_i} . d'_{2_i}}{d_{2_i} + d'_{2_i}} \text{ (mm)}$$

$$\frac{\Delta f_{2\,sys}}{|f_{2\,mid}|} = \left|\frac{1}{d_{2\,mid}} - \frac{1}{d_{2\,mid} + d'_{2\,mid}}\right| \Delta d_{2\,mid} + \left|\frac{1}{d'_{2\,mid}} - \frac{1}{d_{2\,mid} + d'_{2\,mid}}\right| \Delta d'_{2\,mid} \text{ (mm)}$$

$$\Delta f_2 = \underline{\Delta f_2} + \Delta f_{2\,sys} \text{ (mm)}$$

VI/ EXPERIMENTAL DATA

For the convergent lens:

Replication	Silberman's method	Bessel's method

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	L_o (mm)	f_1 (mm)	$\otimes f_1$ (mm)	L (mm)	a (mm)	f_1 (mm)	$\otimes f_1$ (mm)
1	400	100	1,333	450	155	99,153	0,721
2	396	99	0,333	470	192	97,891	0,541
3	388	97	1,667	490	218	98,253	0,179
Mean	394,667	98,667	1,111	470	188,333	98,432	0,480

For the divergent lens:

Replication	Connecting point method			
	$d_2(mm)$	$d_2'(mm)$	$f_2(mm)$	$\Delta f_2(mm)$
1	-50	125	-83,333	3,506
2	-55	202	-75,578	4,249
3	-60	235	-80,571	0,744
Mean	-55	187,33	-79,827	2,833

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VII/ CALCULATIONS

❖ **Silberman's method:**

$f_{1_1} = \frac{L_{o1}}{4} = \frac{400}{4} = 100\ (mm)$

$f_{1_2} = \frac{L_{o2}}{4} = \frac{396}{4} = 99\ (mm)$

$f_{1_3} = \frac{L_{o3}}{4} = \frac{388}{4} = 97\ (mm)$

$\Delta f_{1\,sys} = \frac{\Delta L_{o\,sys}}{4} = \frac{1}{4} = 0{,}25\ (mm)$

$\Delta f_1 = \underline{\Delta f_1} + \Delta f_{1\,sys} = 1{,}111 + 0{,}25 = 1{,}361\ (mm)$

❖ **Bessel's method:**

$f_{1_1} = \frac{{L_1}^2 - {a_1}^2}{4L_1} = \frac{450^2 - 155^2}{4.450} = 99{,}153\ (mm)$

$$f_{1_2} = \frac{{L_2}^2 - {a_2}^2}{4L_2} = \frac{470^2 - 192^2}{4.470} = 97{,}891\ (mm)$$

$$f_{1_3} = \frac{{L_3}^2 - {a_3}^2}{4L_3} = \frac{490^2 - 218^2}{4.490} = 98{,}253\ (mm)$$

$$\frac{\Delta f_{1\,sys}}{f_{1\,mid}} = \left|\frac{2L_{mid}}{L_{mid}^2 - a_{mid}^2} - \frac{1}{L_{mid}}\right| \Delta L_{sys} + \left|\frac{-2a_{mid}}{L_{mid}^2 - a_{mid}^2}\right| \Delta a_{sys}$$

$$= \left|\frac{2.470}{470^2 - 192^2} - \frac{1}{470}\right| . 1 + \left|\frac{-2.192}{470^2 - 192^2}\right| . 1$$

$=0{,}005$

$$\Rightarrow \Delta f_{1\,sys} = 0{,}005\,.\,f_{1\,mid} = 0{,}005\,.97{,}891 = 0{,}489\ (mm)$$

$\Delta f_1 = \underline{\Delta f_1} + \Delta f_{1\,sys} = 0{,}480 + 0{,}489 = 0{,}969$ (mm)

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❖ **Connecting point method:**

$$f_{2_1} = \frac{d_{2_1}.d'_{2_1}}{d_{2_1} + d'_{2_1}} = \frac{(-50).125}{(-50) + 125} = -83{,}333\ (mm)$$

$$f_{2_2} = \frac{d_{2_2}.d'_{2_2}}{d_{2_2} + d'_{2_2}} = \frac{(-55).202}{(-55) + 202} = -75{,}578\ (mm)$$

$$f_{2_3} = \frac{d_{2_3}.d'_{2_3}}{d_{2_3} + d'_{2_3}} = \frac{(-60).235}{(-60) + 235} = -80{,}571\ (mm)$$

$$\frac{\Delta f_{2\,sys}}{|f_{2\,mid}|} = \left|\frac{1}{d_{2\,mid}} - \frac{1}{d_{2\,mid} + d'_{2\,mid}}\right| \Delta d_{2\,sys} + \left|\frac{1}{d'_{2\,mid}} - \frac{1}{d_{2\,mid} + d'_{2\,mid}}\right| \Delta d'_{2\,sys}$$

$$= \left|\frac{1}{-55} - \frac{1}{-55 + 202}\right| .1 + \left|\frac{1}{202} - \frac{1}{-55 + 202}\right| .1$$

$$=0{,}0268$$

$$\Rightarrow \Delta f_{2\,sys} = 0{,}0268 \,.\, f_{2\,mid} = 0{,}0268 \,.\, 75{,}578 = 2{,}025\ (mm)$$

$$\Delta f_2 = \underline{\Delta f_2} + \Delta f_{2\,sys} = 2{,}833 + 2{,}025 = 4{,}858\ (mm)$$

VIII/ CONCLUSION

- **Silberman's method:** $f_1 = \underline{f_1} \pm \Delta f_1 = 98{,}667 \pm 1{,}361\ (mm)$
- **Bessel's method:** $f_1 = \underline{f_1} \pm \Delta f_1 = 98{,}432 \pm 0{,}969\ (mm)$
- **Connecting point method:** $f_2 = \underline{f_2} \pm \Delta f_2 = -79{,}827 \pm 4{,}858\ (mm)$

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